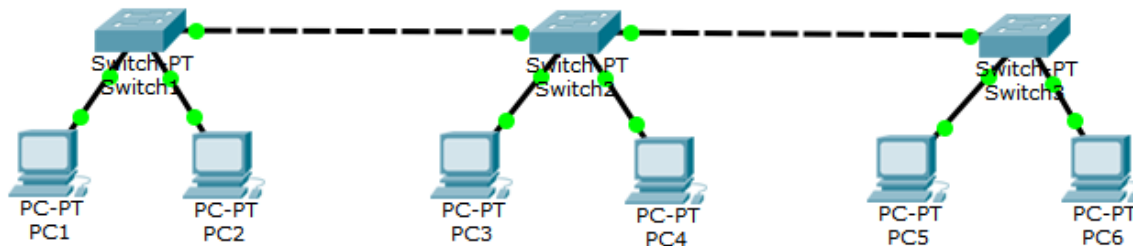




Lab 3 – Study of Different VLAN Communication Cases on Cisco Switches

We will use **Cisco Packet Tracer** to configure VLANs on Cisco switches.



Objective: Study the different possibilities of communication between VLANs.

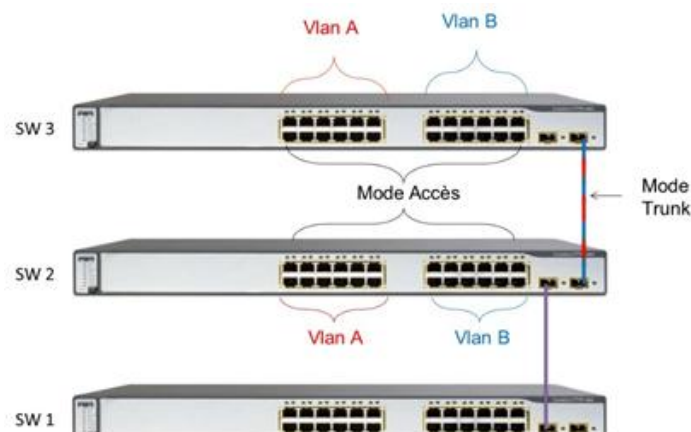
Step 1: Creating the topology and testing machine connectivity

- Build a topology using 3 switches and 6 computers.
- All 6 computers belong to the same network: 192.168.1.0/24.
- Assign IP addresses from 192.168.1.1/24 to 192.168.1.6/24.
- Test connectivity (ping) and make sure all computers can communicate with each other.
- Record the ping results.

Step 2: Creating VLANs and assigning interfaces to VLANs

Case 1:

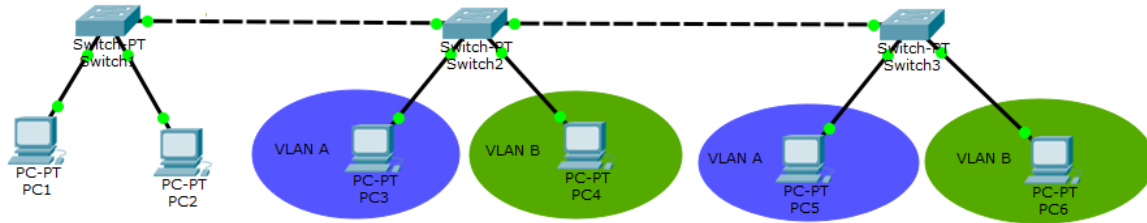
- On SW1, no VLANs are configured.
- On SW2, create 2 VLANs:
 - PC3 in VLAN 2 (name: A)
 - PC4 in VLAN 3 (name: B)
- On SW3, do the same:
 - PC5 in VLAN A
 - PC6 in VLAN B



Commands to use:

1. Create VLANs with the vlan command.
2. Verify creation with do show vlan.
3. Assign interfaces to VLANs using the switchport command.
4. Re-check with do show vlan.

Reminder: If needed, refer back to Lab 1 for details on VLAN creation and interface assignment.



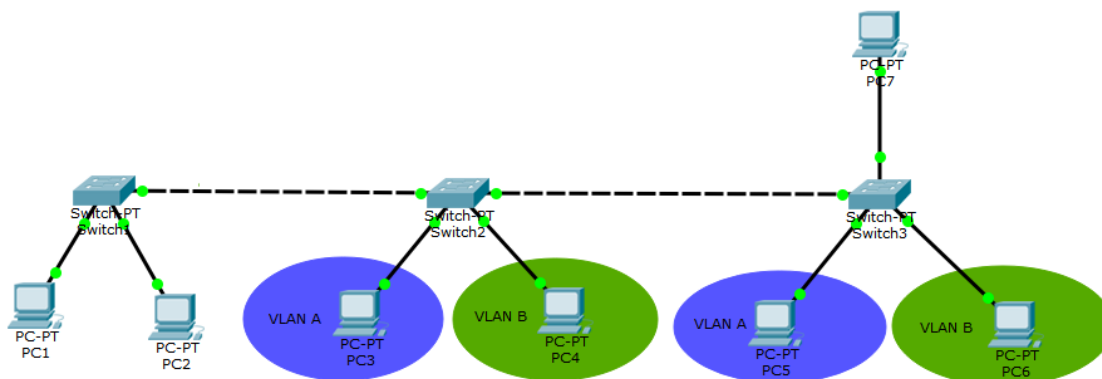
Test communication (ping) between different machines and note results.

Expected Results:

- PCs in the same VLAN but on different switches cannot communicate initially.
- Configure a **trunk link** between SW2 and SW3.
- Test communications again.

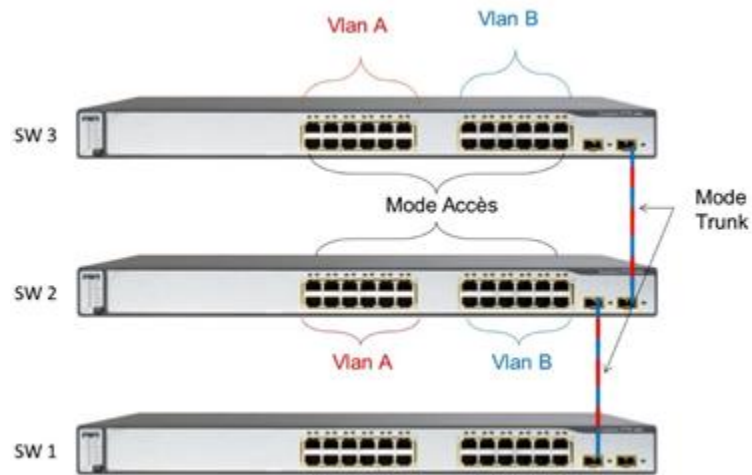
Expected Results:

- Since SW2 and SW3 are connected via a trunk, VLAN A ports can now communicate (PC3 ↔ PC5) and VLAN B ports can communicate (PC4 ↔ PC6).
- No communication is possible with PCs on SW1, since they are not in VLAN A or B.
- Add **PC7 (192.168.1.7)** to SW3 in VLAN 1, then re-test. PC7 should be able to communicate with PC1 and PC2 on SW1.



Case 2:

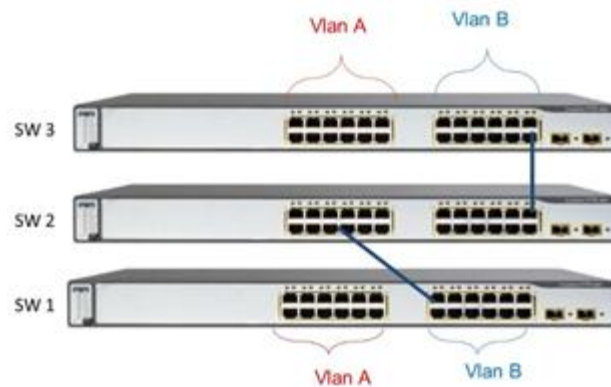
- Configure a trunk between SW1 and SW2.
- Run connectivity tests and record results.



Expected Results:

- Same as Case 1.

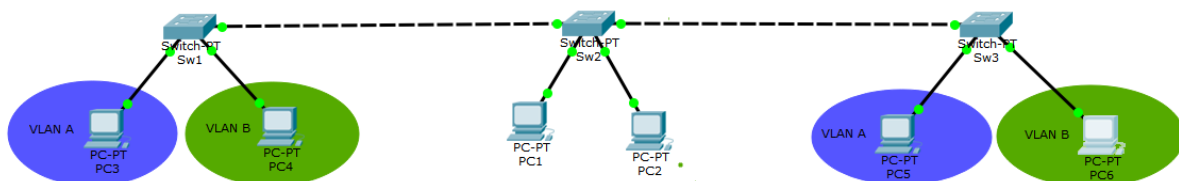
Case 3:



- On **SW1**, add interface Fa2/1 to VLAN B.

VLAN	Name	Status	Ports
1	default	active	Fa3/1, Fa4/1, Fa5/1
2	A	active	Fa1/1
3	B	active	Fa0/1, Fa2/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

- Do the same on **SW3**.
- Create direct connections:
 - SW1's Fa2/1 to a port on SW2
 - SW3's Fa2/1 to a port on SW2



- Test and record results.

Expected Results:

- PCs in VLAN B can communicate with each other and with PCs on SW2.
- PCs in VLAN A on SW1 and SW3 cannot communicate with each other nor with PCs on SW2.

Reminder: If using the same topology as in previous cases, make sure to return trunk ports to VLAN 1 before reconfiguring.

Final Note:

Try experimenting with other cases studied in class to deepen your understanding.